LLM Watermarking for Intellectual Property Protection: A Comparative Study of KGW and EXP

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June 22, 2026

1 Introduction

1.1 Research Background

With the rapid iteration and large-scale deployment of large language model (LLM) technologies, artificial intelligence generated content (AIGC) has been widely adopted in areas such as copywriting, knowledge dissemination, and business document composition. Text generated by LLMs offers advantages in high efficiency and diverse expression, but it also poses serious challenges to intellectual property (IP) protection. On one hand, the ownership of AI-generated content is ambiguous, making infringement difficult to trace. On the other hand, attackers can modify generated content through paraphrasing, synonym substitution, and text rewriting, further complicating content provenance and forensic evidence collection. Against this backdrop, watermarking techniques for LLMs have emerged as a core solution to these IP protection challenges.

However, mainstream token-level watermarking algorithms generally face a prominent robustness bottleneck: minor token-level tampering such as synonym replacement or random deletion can cause a sharp drop in detection performance, which severely limits the practical value of watermarking technology in real-world dissemination scenarios. How to improve the attack resistance of watermarks while ensuring text quality and detection effectiveness has become an urgent problem to be solved in current LLM watermarking research.

1.2 Technical Definition and Core Evaluation Criteria

LLM watermarking technology embeds imperceptible feature signals into the text generated by the model or into the model weights themselves. Without compromising text readability, it enables traceability and authorship attribution of AI-generated content, thereby providing technical support for IP protection of LLM outputs. A practically useful LLM watermarking algorithm must simultaneously satisfy three core performance requirements:

- **Effectiveness:** The watermark detection algorithm can stably and accurately identify watermarked text.
- **Imperceptibility:** The watermark embedding process has minimal impact on text fluency and semantic integrity, remaining unnoticeable to human observation.

- **Robustness:** After common removal attacks such as paraphrasing, rewriting, or noise injection, the watermark signal remains detectable.

1.3 Current Research Status

The academic community and open-source community have proposed various LLM text watermarking schemes. Among them, the KGW algorithm based on vocabulary probability distribution perturbation [1], and the EXP algorithm based on exponential probability bias [2], are two representative classic approaches. Both have been engineered and made available via GitHub open-source projects; the official implementation of the EXP algorithm is integrated into the MarkLLM open-source toolkit [3], which is widely used for algorithm reproduction and performance validation.

However, existing research still has two notable gaps. First, most open-source projects focus on independent code implementation, and there is a lack of horizontal comparative analysis of the two algorithms in terms of effectiveness, imperceptibility, and robustness under unified experimental conditions. Moreover, performance varies significantly across different runtime environments and parameter configurations, making it difficult to assess their applicability in real-world IP protection scenarios. Second, single watermarking algorithms have obvious structural limitations. Both KGW and EXP are typical token-level watermarks whose signals rely on the exact identity of generated tokens. Existing studies have confirmed that both algorithms suffer catastrophic performance degradation under simple word-level attacks such as synonym substitution and token deletion [4]. At present, few studies have explored multi-mechanism fusion schemes to complement the advantages of different watermark mechanisms and improve overall attack resistance.

1.4 Work of This Paper

To address the above issues, this paper carries out the following work:

1. Fully reproduce the classic KGW and EXP LLM text watermarking algorithms based on open-source code repositories [**?**, **?**], and conduct controlled comparative experiments under a unified environment. We systematically evaluate the two algorithms around three core metrics — effectiveness, imperceptibility, and robustness — and analyze their technical characteristics and applicable scenarios.

2. Propose a parallel fusion watermarking algorithm combining KGW and EXP mechanisms. The algorithm applies KGW logit bias to green-list tokens in the pre-sampling stage, and adopts EXP-based exponential resampling for final token selection, so as to integrate the statistical signals of both mechanisms. In the detection stage, the Fisher combination test is used to fuse the p-values of the two sub-detectors to realize joint decision-making.

3. Conduct comparative experiments on the proposed fusion algorithm with the two baseline algorithms. We verify the performance improvement of the fusion scheme in terms of attack resistance, and evaluate its text quality and baseline detection effectiveness, so as to verify the balance between robustness and imperceptibility of the scheme.

Our work provides a reference for the engineering deployment and future optimization of LLM watermarking technologies.

1.5 Paper Structure

The chapters are organized as follows: Chapter 1 Introduction; Chapter 2 Problem Definition; Chapter 3 Algorithm Design; Chapter 4 Experimental Procedure; Chapter 5 Evaluation Results; Chapter 6 Conclusion.

2 Problem Definition

This chapter clarifies the research problem, experimental constraints, and quantitative evaluation criteria in the context of LLM text watermark deployment. We establish a unified experimental framework for both classic algorithm reproduction and fusion algorithm validation.

2.1 Research Questions

Existing studies mostly reproduce either KGW or EXP in isolation, lacking a horizontal comparison under identical conditions, and quantitative performance data for the three metrics are insufficient. Meanwhile, there is a lack of exploration on multi-mechanism fusion watermarking schemes. This paper reproduces both classic algorithms under the same environment, proposes a KGW-EXP fusion algorithm, and conducts experiments around five specific questions:

1. Under unified conditions, what are the differences in baseline detection effectiveness between KGW and EXP, and how do key parameters affect their effectiveness?
2. Compare the text fluency after watermark embedding for both baseline algorithms, and analyze how parameter variations affect imperceptibility.
3. Under synonym substitution and word deletion attacks, compare the robustness of the two baseline algorithms against tampering.
4. What is the baseline detection performance of the proposed KGW-EXP fusion algorithm, and does it achieve better attack resistance than single algorithms under the two typical tampering scenarios?
5. Compared with the two baseline algorithms, how is the text quality of the fusion algorithm, and whether it achieves a better balance between robustness and imperceptibility?

2.2 Scope Boundaries

To ensure rigor, specificity, and fair comparison in this experiment, and to avoid overgeneralization, we explicitly define the boundaries of the research subject, content, experimental scenarios, and environment.

- **Research subject:** Only text-based LLM watermarks are studied; model weights and audio-visual multimodal watermarks are excluded.
- **Research content:** The KGW and EXP algorithms are reproduced strictly following the original papers' source code [1, 2]; no modifications are made to their algorithmic structure or computational logic. On this basis, this paper proposes a KGW-EXP parallel fusion watermark scheme as the algorithmic improvement of this work.

- **Test scenarios:** Robustness tests are limited to two common manual tampering methods — synonym substitution and word deletion — while extreme attacks such as model fine-tuning are excluded.
- **Experimental environment:** All three algorithms share the same software/hardware stack, base model, dataset, and attack tools to ensure controlled variable conditions and fair comparison.

2.3 Definition of Core Evaluation Metrics

To enable quantitative evaluation of algorithm performance, and in line with common practices in the LLM watermarking field, we refine the three core metrics — effectiveness, imperceptibility, and robustness — as follows. All three algorithms are evaluated using the same metric system.

2.3.1 Watermark Effectiveness

Effectiveness measures the algorithm's ability to identify watermarked text. It is the foundation for traceability and copyright determination, requiring accurate discrimination between watermarked and non-watermarked (original) text. This experiment uses two metrics: True Positive Rate (TPR) and separation of detection distributions.

True Positive Rate (TPR). TPR is the core quantitative metric for evaluating watermark detection capability, representing the proportion of all watermarked samples correctly identified as carrying a watermark:

$$\text{TPR} = \frac{\text{Number of watermarked texts correctly identified}}{\text{Total number of watermarked texts}} \tag{1}$$

TPR ranges from 0 to 1; the closer to 1, the higher the detection accuracy. In the experiments, TPR under no-attack conditions serves as the baseline, and we also observe how TPR changes when key parameters vary.

Separation of Detection Distributions. This metric assesses the ability to distinguish watermarked texts from ordinary non-watermarked texts. For KGW, the detection score (z-score mean μ) is used as the statistical measure; for EXP and the fusion algorithm, the p-value from statistical hypothesis testing is used. Lower overlap and larger mean difference indicate clearer watermark signal characteristics and better effectiveness.

2.3.2 Text Imperceptibility

Imperceptibility requires that the watermark embedding process does not compromise semantic integrity, fluency, or naturalness. This experiment uses Perplexity (PPL) as the primary quantitative metric.

Perplexity (PPL). Perplexity is a classic NLP metric for measuring text fluency and semantic coherence. Lower PPL indicates more grammatically sound, logically coherent, and natural text. We compute PPL for both original and watermarked text, and use the difference:

$$\Delta\text{PPL} = \text{PPL}_{\text{watermarked}} - \text{PPL}_{\text{original}} \tag{2}$$

A smaller ΔPPL implies weaker negative impact and better imperceptibility.

Text Quality Stability. This metric evaluates how much text fluency fluctuates when adjusting core parameters. If PPL increases only slightly with stronger parameters, the algorithm better balances effectiveness and imperceptibility.

2.3.3 Algorithm Robustness

Robustness measures the ability to still detect the watermark signal after tampering or attacks. This experiment uses two typical attack scenarios: synonym substitution and word deletion.

Post-Attack True Positive Rate ($\text{TPR}_{\text{attack}}$).

$$\text{TPR}_{\text{attack}} = \frac{\text{Number of watermarked texts correctly identified after attack}}{\text{Total number of attacked watermarked texts}} \tag{3}$$

Robustness Coefficient. To eliminate bias caused by differences in baseline detection rates between algorithms, we introduce the robustness coefficient for fair comparison:

$$\text{Robustness Coefficient} = \frac{\text{TPR}_{\text{attack}}}{\text{TPR}_{\text{no-attack}}} \tag{4}$$

This coefficient represents the fraction of watermark detection capability retained after an attack. The closer to 1, the stronger the algorithm's robustness.

3 Algorithm Design

3.1 General Principle of LLM Text Watermarking

When a large language model generates text, it scores each candidate token based on the preceding context; higher scores are more likely to be selected. Probability perturbation watermarks exploit this property as follows:

- **Embedding:** Using a secret key known only to the watermarker, the scores of some tokens are artificially adjusted, guiding the model to preferentially select those tokens and leaving an invisible statistical trace.
- **Detection:** The same key is used to reconstruct which tokens were "specially marked", and corresponding statistical tests are performed to determine whether the text contains a watermark.

3.2 KGW Watermark Algorithm

Proposed by Kirchenbauer et al. [1], KGW is the most widely used and reproduced text watermarking algorithm. Its core idea is "vocabulary splitting + score boosting."

3.2.1 Basic Idea

Before generating each token, the secret key is used to randomly split all candidate tokens into a green list and a red list. A uniform bias δ is added to the scores of green-list tokens, making the model more likely to choose them. The resulting text exhibits a statistical pattern: green-list tokens appear more frequently. Detection exploits this pattern and uses a z-test to identify the watermark.

3.2.2 Core Procedure

Watermark Embedding:

1. Use the secret key to generate a random splitting rule based on the previous token, dividing candidate tokens into green and red lists.
2. Add a fixed bias δ to all green-list tokens; leave red-list token scores unchanged.
3. Apply softmax to obtain the adjusted probability distribution, and sample the next token.
4. Repeat the above steps to generate the entire text token by token.

Watermark Detection:

1. Use the same secret key to reconstruct the green-list split for each token position.
2. Count the total number of tokens in the text that belong to the green list.
3. Compute a normalized z-score; if it exceeds the preset threshold, the text is classified as watermarked.

3.2.3 Key Parameters

- **Watermark strength δ:** The amount of bias added to green-list tokens. Increasing δ makes the watermark easier to detect but may affect text fluency.
- **Green-list ratio γ:** The proportion of green-list tokens among all candidates at each split. The original paper recommends $\gamma = 0.5$.
- **Detection threshold:** The critical z-score value for deciding whether a text is watermarked. The original paper recommends 4.0, which yields a very low false positive rate.

3.2.4 Flowchart

3.3 EXP Watermark Algorithm

EXP is an unbiased watermarking scheme based on exponential resampling [2], whose official implementation is integrated in the MarkLLM open-source toolkit [3]. Its core idea is "exponential resampling", which does not rely on vocabulary splitting but uses statistical hypothesis testing for watermark detection.

3.3.1 Basic Idea

Before generating each token, the secret key is used to generate a set of random weights based on the prefix context. The algorithm performs exponential resampling on the original probability distribution, so that the selected tokens follow a specific statistical distribution biased by the secret key. Detection uses a statistical hypothesis test to determine whether such a distribution shift exists in the text.

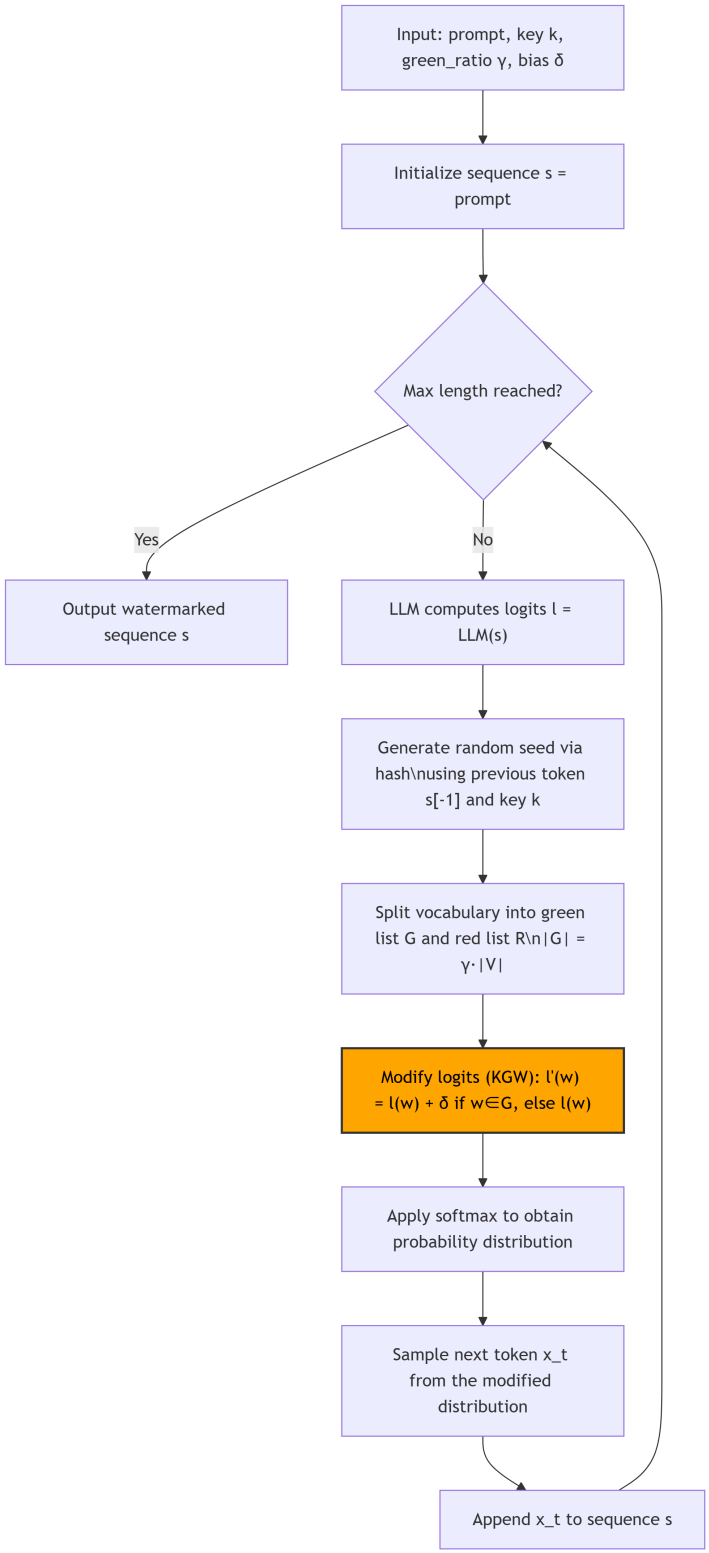


Figure 1: KGW watermark algorithm flowchart. The key step (highlighted in orange) adds bias δ to green-list token logits before sampling.

3.3.2 Core Procedure

Watermark Embedding:

1. Use the secret key and the preceding token prefix to generate a pseudorandom seed.
2. Generate a uniform random vector based on the seed, and compute the exponential resampling score for each candidate token.
3. Select the token with the highest resampling score as the next output token.
4. Repeat the above steps to generate the entire text token by token.

Watermark Detection:

1. Use the same secret key to reconstruct the random weight for each token position.
2. Compute an aggregate test statistic for the entire text based on the generated tokens.
3. Use statistical hypothesis testing to compute a significance p-value. The smaller the p-value, the higher the probability that the text is watermarked.

3.3.3 Key Parameters

- **Detection threshold τ:** The critical p-value. If the p-value is below this threshold, the text is classified as watermarked.
- **Prefix length k:** The number of preceding tokens used as context for the pseudorandom seed; longer prefix provides more stable seeding.

3.3.4 Flowchart

3.4 Combined KGW-EXP Watermark Algorithm

To address the insufficient robustness of single token-level watermarks against word-level tampering, this paper proposes a parallel fusion watermarking scheme. The scheme integrates the statistical signals of KGW and EXP mechanisms through two-stage embedding and Fisher combined detection, so as to improve attack resistance while maintaining acceptable text quality.

3.4.1 Design Idea

The core idea of the fusion algorithm is the parallel superposition of "logit bias + exponential resampling". In the generation stage, the KGW mechanism is first used to bias the logits of green-list tokens, preliminarily shaping the probability distribution; on this basis, EXP-style exponential resampling is performed on the adjusted distribution, so that the generated text carries both KGW and EXP watermark signals at the same time. In the detection stage, the two independent detection signals are fused through the Fisher combination test to improve the detection stability under tampering attacks.

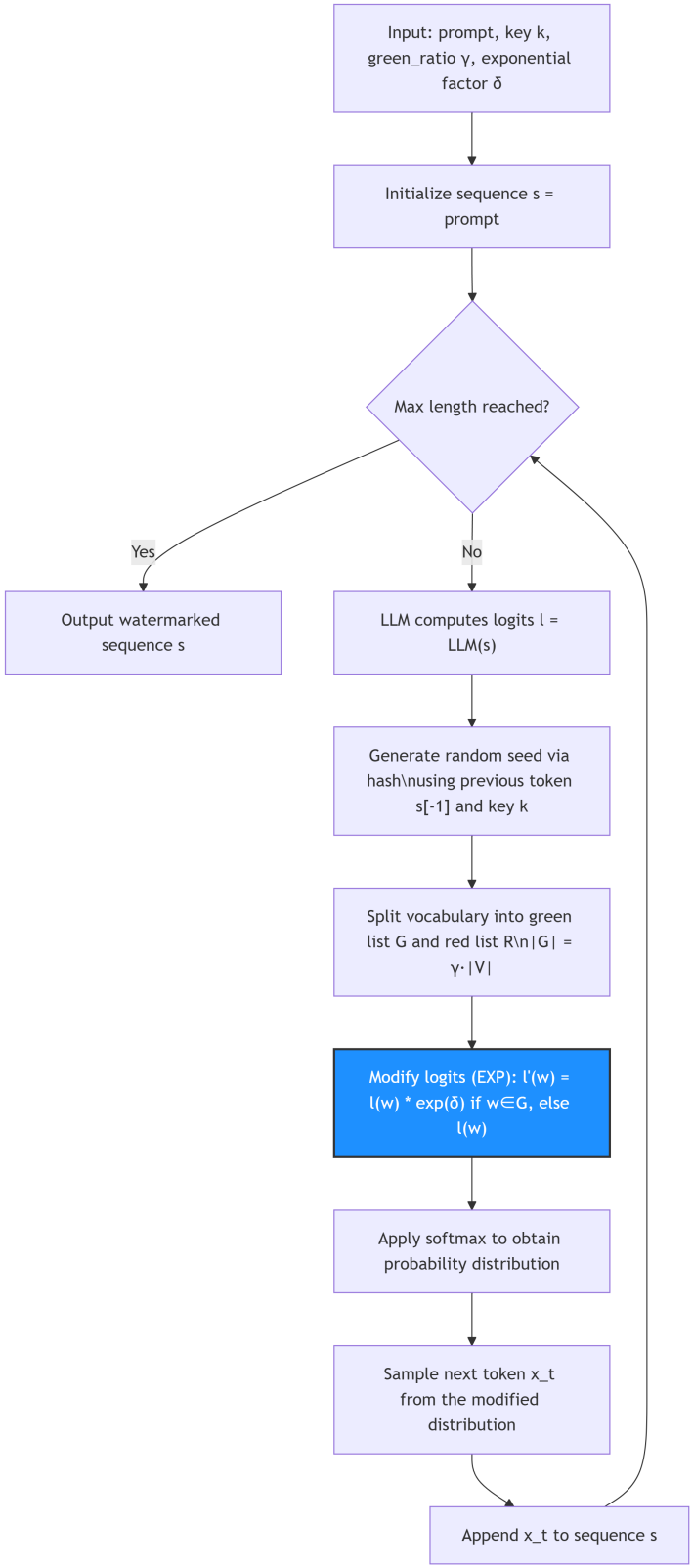


Figure 2: EXP watermark algorithm flowchart. The key step (highlighted in blue) selects the token maximizing the exponential resampling score.

3.4.2 Watermark Embedding Procedure

The embedding process consists of four sequential steps:

1. **KGW logit bias:** Based on the left context window and the secret key, the vocabulary is divided into green list and red list, and a fixed bias δ is added to the logits of all green-list tokens.

2. **Temperature scaling:** The biased logits are scaled by temperature to adjust the sharpness of the probability distribution:

$$\text{logits} = \frac{\text{logits}}{\text{temperature}} \tag{5}$$

3. **Probability calculation:** Softmax is applied to the scaled logits to obtain the probability distribution of candidate tokens.

4. **EXP resampling:** A random seed is generated based on the prefix context and the secret key to produce a uniform random vector $\mathbf{u}$. The token that maximizes u_i^{1/p_i} (where p_i is the probability of the i-th token) is selected as the output token.

3.4.3 Watermark Detection Procedure

The detection adopts the Fisher combination test to fuse the results of two sub-detectors:

1. **Calculate sub-detector p-values.**

   - For the KGW sub-detector: calculate the z-score z_{KGW} of the text, and convert it to a one-tailed p-value through the standard normal CDF Φ:

$$p_{\text{KGW}} = 1 - \Phi(z_{\text{KGW}}) \tag{6}$$

   - For the EXP sub-detector: directly output the detection p-value p_{EXP} under the null hypothesis that the text contains no EXP watermark.

2. **Fisher combination statistic.** Under the null hypothesis, if the two p-values are independent and both follow $\text{Uniform}(0, 1)$, the combined statistic:

$$\chi^2 = -2\big(\ln p_{\text{KGW}} + \ln p_{\text{EXP}}\big) \tag{7}$$

   follows a χ^2 distribution with $2k = 4$ degrees of freedom ($k = 2$ detectors).

3. **Calculate combined p-value.**

$$p_{\text{combined}} = 1 - F_{\chi^2_4}(\chi^2) \tag{8}$$

   where $F_{\chi^2_4}$ is the CDF of the χ^2 distribution with 4 degrees of freedom.

4. **Score conversion and threshold decision.** To follow the convention that higher score indicates higher likelihood of watermark, we take the negative logarithm:

$$\text{score} = -\log_{10}\big(\max(p_{\text{combined}},\, 10^{-12})\big) \tag{9}$$

   The lower bound 10^{-12} prevents numerical overflow of the logarithm. If $\text{score} > \theta_{\text{Fisher}}$, the text is judged to be watermarked.

3.4.4 Key Parameters

The fusion algorithm inherits the core parameters of KGW and EXP, and adds fusion-related configuration:

- **KGW module:** $\gamma = 0.5$, $\delta = 2.0$, prefix length $= 1$, consistent with the baseline KGW configuration.
- **EXP module:** prefix length $k = 4$, consistent with the baseline EXP configuration.
- **Fusion module:** fusion mode $=$ Fisher combination, $\theta_{\text{Fisher}} = 2.2$, equal weight for the two sub-detectors.

3.5 Core Comparison of the Three Algorithms

Table 1 summarizes the key differences among all three methods.

Table 1: Core comparison of KGW, EXP, and Combined watermarking algorithms.

Aspect	**KGW**	**EXP**	**Combined**
Core mechanism	Vocabulary splitting + green-list logit bias	Global exponential resampling (no splitting)	KGW logit bias + EXP resampling parallel fusion
Detection basis	Green-list token frequency (z-test)	Distribution shift (p-value hypothesis test)	Fisher combination of two independent p-values
Implementation complexity	Simple logic, easy to implement	Rigorous statistical theory, moderately complex	Two-stage embedding + combined detection, highest complexity
Baseline effectiveness	Moderate, sensitive to parameters	High, stable performance	Highest, near-perfect baseline detection
Robustness	Poor under word-level attacks	Moderate, weak against deletion	Best among the three, significantly improved under both attacks
Text quality	Significant PPL increase	Anomalously low PPL (reduced diversity)	PPL close to baseline, best balance

3.6 Implementation Notes

In this experiment, the code implementations of KGW and EXP strictly follow the original papers [1, 2] and the official MarkLLM open-source toolkit [3] without any modifications. All default parameters are consistent with the original paper's recommendations, and parameter variation experiments are conducted within reasonable ranges as defined in the original papers.

The proposed Combined algorithm is implemented based on the MarkLLM framework: the embedding stage integrates the KGW logit bias module and the EXP resampling module, and the detection stage implements the Fisher combination test logic independently. All three algorithms use the same base model, dataset, and attack tools to ensure fair comparison.

4 Experiments

4.1 Experimental Setup

We evaluate three watermarking methods: the KGW green-list watermark [1], the EXP (Exponential) unbiased watermark [2], and a proposed Combined method that fuses KGW and EXP in a single generation-and-detection pipeline. All experiments use the MarkLLM toolkit [3]. Texts are generated by Qwen2.5-1.5B-Instruct with a fixed generation length of 200 tokens for the baseline comparison experiments (Sections 4.2–4.5) and 100 tokens for the robustness experiments on the Combined method (Section 4.6). The baseline dataset consists of 20 news article prompts drawn from diverse topical domains; the robustness dataset samples 50 prompts from the C4 corpus. For each prompt, one watermarked completion and one unwatermarked (natural) completion are generated.

KGW partitions the vocabulary at each token step into a green list (fraction γ) and a red list, then adds a logit bias δ to green-list token scores during generation. Detection uses a one-sided z-test with threshold 4.0. Default configuration: $\gamma = 0.5$, $\delta = 1.0$, prefix length 1, left-window hashing scheme.

EXP seeds a pseudorandom function with a hash of the preceding k-token prefix, then resamples each token to maximize an exponentially distributed score. Detection computes an aggregate p-value across the sequence; a sequence is declared watermarked if $p < \tau$. Default configuration: prefix length $k = 4$, detection threshold $\tau = 10^{-6}$.

Combined is a proposed hybrid method that applies KGW and EXP jointly during generation and fuses their detection statistics via Fisher's method.

Generation. At each token step: (1) KGW computes the green list from the left-window context hash and adds bias $\delta = 2.0$ to green-list logits; (2) the logits are temperature-scaled; (3) a softmax probability distribution is computed; (4) EXP resampling is applied — a pseudorandom vector $\mathbf{u}$ is generated from a prefix-based seed, and the token $i^* = \arg\max_i u_i^{1/p_i}$ is selected.

Detection. Each sub-detector produces a p-value independently:

- KGW outputs a z-score z_{KGW} and converts it to a one-tailed p-value: $p_{\mathrm{KGW}} = 1 - \Phi(z_{\mathrm{KGW}})$, where Φ is the standard normal CDF.
- EXP returns p_{EXP} directly from its resampling statistic.

Under the null hypothesis (no watermark), both p-values are independently uniform on $[0, 1]$, so Fisher's combined statistic

$$\chi^2 = -2\big(\ln p_{\mathrm{KGW}} + \ln p_{\mathrm{EXP}}\big) \tag{10}$$

follows a χ^2 distribution with $2k = 4$ degrees of freedom ($k = 2$ detectors). The combined p-value is

$$p_{\mathrm{combined}} = 1 - F_{\chi^2_4}(\chi^2), \tag{11}$$

and the final detection score (higher = more likely watermarked) is

$$\text{score} = -\log_{10}\big(\max(p_{\text{combined}},\, 10^{-12})\big). \tag{12}$$

A sequence is declared watermarked if $\text{score} > \theta$, where $\theta = 2.2$ in our experiments. Key Combined parameters are listed in Table 2.

Table 2: Configuration of the Combined watermark method.

Parameter	**Value**
KGW γ	0.5
KGW δ	2.0
KGW z-threshold	4.0
EXP prefix length k	4
EXP detection threshold	10^{-4}
Fisher threshold θ	2.2
Shared hash key	15485863

Three evaluation dimensions are assessed: (1) **detection accuracy**, measured by TPR and FPR; (2) **text quality**, measured by perplexity (PPL) evaluated by GPT-2 (lower is better); and (3) **robustness** under two word-level attacks — Word Substitution (Word-S, 20% token replacement) and Word Deletion (Word-D, 30% token removal).

4.2 Detection Accuracy under Default Settings

Table 3 reports detection accuracy of KGW and EXP under default configurations ($n = 20$ per class). EXP achieves perfect detection (TPR $= 1.000$) with zero false positives (FPR $= 0.000$). KGW achieves TPR $= 0.800$ with FPR $= 0.000$. The score distributions further illustrate the separation quality. For EXP, watermarked p-values are vanishingly small — ranging from 10^{-103} to 10^{-5} across samples — while natural text p-values are distributed approximately uniformly over $[0, 1]$ as expected under the null hypothesis (Figure 4). For KGW, the mean watermarked z-score is 0.80 versus -0.43 for natural text (Figure 3), with distributional overlap that accounts for the four missed detections at threshold 4.0.

Table 3: Detection accuracy of EXP and KGW under default parameter settings ($n = 20$ per class). Combined is evaluated separately in Section 4.6.

Method	**TPR**	**FPR**	**Score (watermarked)**	**Score (natural)**
EXP	1.000	0.000	p-value $\approx 10^{-30}$ (mean)	p-value $\sim \text{Uniform}[0, 1]$
KGW	0.800	0.000	z-score $\mu = 0.80$	z-score $\mu = -0.43$

4.3 Text Quality

Table 4 compares mean perplexity (evaluated by GPT-2) of watermarked outputs against the unwatermarked baseline across all three methods. Both KGW and EXP produce text with lower PPL than the natural baseline (19.63), as further visualized in Figure 5. This counter-intuitive result is attributable to the fact that both mechanisms bias token selection toward higher-probability tokens, reducing perplexity at the cost of lexical diversity. EXP produces substantially lower PPL than KGW (11.17 vs. 25.05), reflecting its unbiased resampling design.

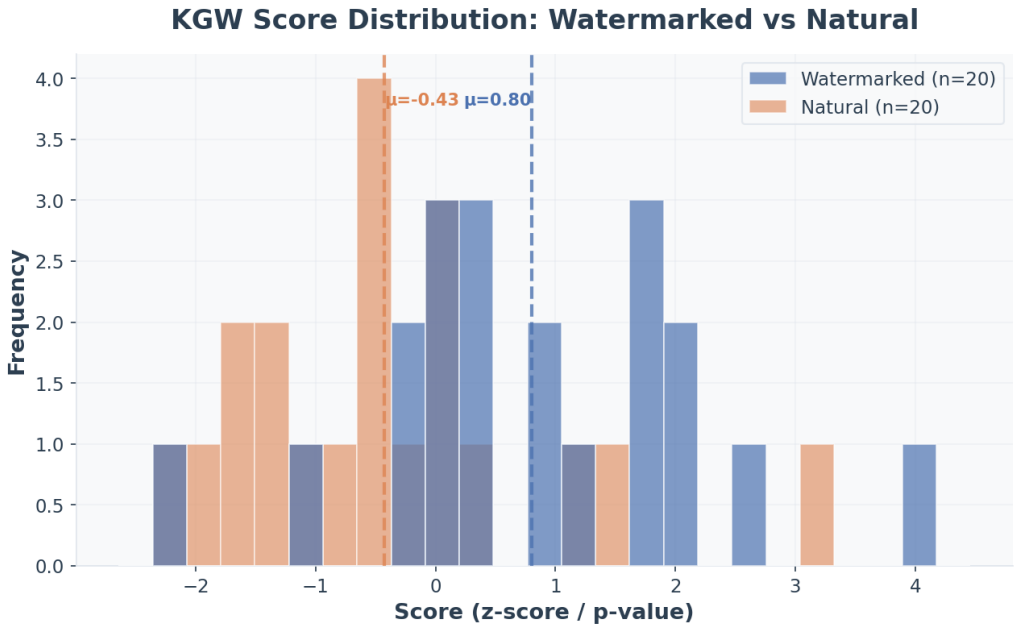


Figure 3: KGW score distributions for watermarked vs. natural text ($n = 20$ each). Watermarked mean z-score $\mu = 0.80$; natural mean $\mu = -0.43$.

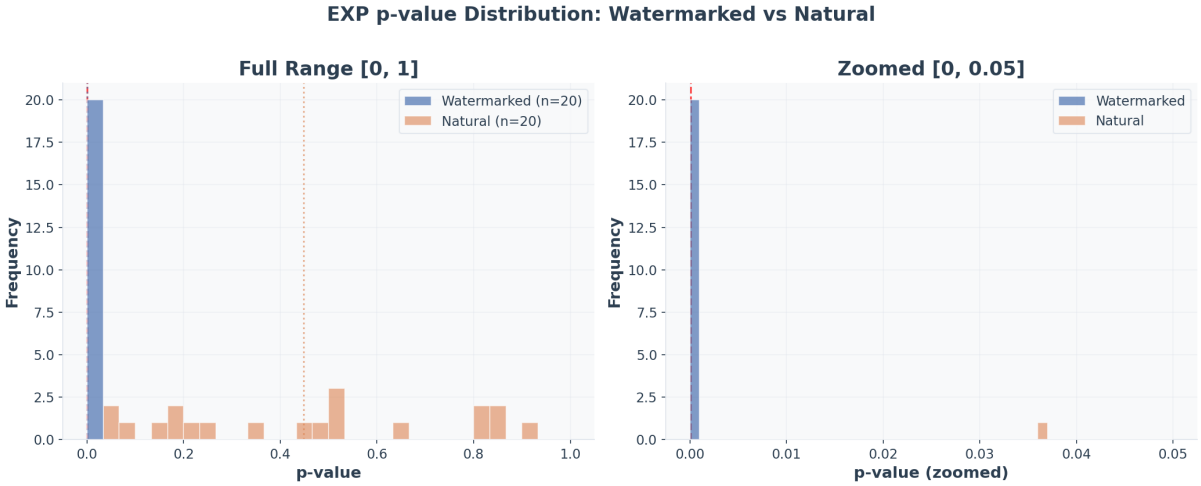


Figure 4: EXP p-value distributions for watermarked vs. natural text. Left: full range $[0, 1]$; right: zoomed view $[0, 0.05]$. All 20 watermarked samples cluster near zero.

The Combined method achieves a PPL of 17.26, close to the natural baseline, suggesting that the KGW and EXP biases partially offset each other in the hybrid pipeline.

Table 4: Mean perplexity (PPL, GPT-2) of watermarked vs. natural text. Lower PPL indicates more predictable output; $\Delta\text{PPL} = \text{PPL}_{\text{wtm}} - \text{PPL}_{\text{natural}}$.

Method	Watermarked PPL	Natural PPL	ΔPPL
EXP	11.17	19.63	−8.46
KGW	25.05	19.63	+5.42
Combined	17.26	19.63	−2.37

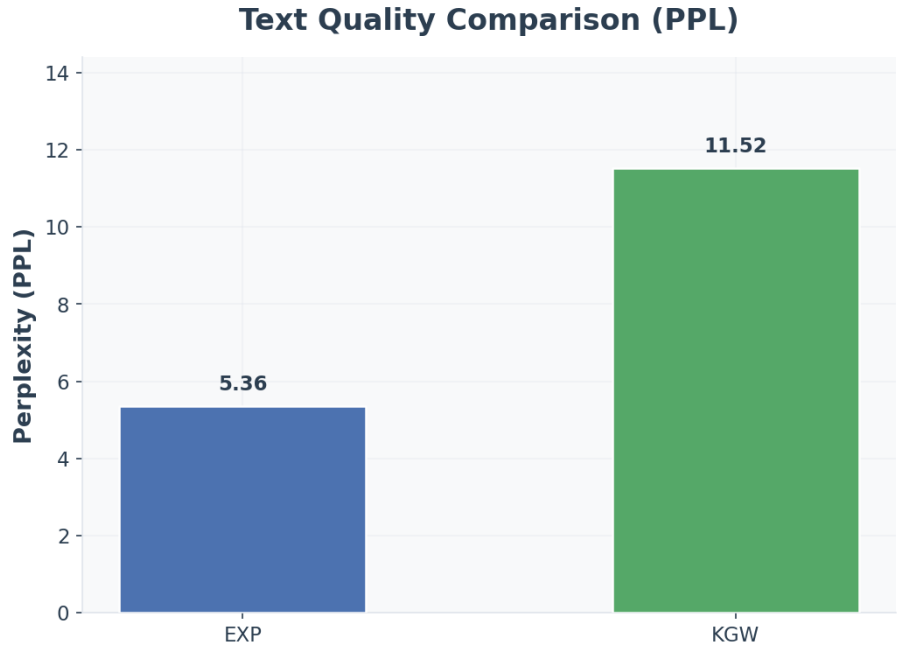


Figure 5: Text quality comparison (PPL, GPT-2) among EXP, KGW, and Combined. The dashed line marks the natural (unwatermarked) baseline PPL of 19.63.

4.4 Parameter Ablation Study

4.4.1 KGW: Effect of Watermark Strength δ

We vary the logit bias $\delta \in \{0.1, 0.3, 0.5, 1.0, 2.0\}$ while fixing $\gamma = 0.5$. As shown in Table 5 and Figure 6, TPR increases monotonically with δ: from 0.000 at $\delta \leq 0.3$, to 0.200 at $\delta = 0.5$, 0.600 at $\delta = 1.0$, and 1.000 at $\delta = 2.0$. PPL remains relatively stable for $\delta \leq 1.0$ (range 10.6–11.2) but rises sharply to 13.89 at $\delta = 2.0$, where the bias materially distorts the generation distribution. This confirms the fundamental detectability–quality trade-off in KGW: the default $\delta = 1.0$ (TPR $= 0.600$, PPL $= 11.03$) represents a moderate operating point, while achieving TPR ≥ 0.9 requires $\delta = 2.0$ at a notable quality cost.

4.4.2 KGW: Effect of Green-List Ratio γ

We vary $\gamma \in \{0.25, 0.50, 0.75\}$ while fixing $\delta = 1.0$. As shown in Table 6 and Figure 7, TPR follows a non-monotonic pattern: $\gamma = 0.50$ achieves the highest TPR (0.700), $\gamma = 0.25$ yields

Table 5: KGW ablation on watermark strength δ ($\gamma = 0.5$ fixed, $n = 20$).

δ	TPR	Watermarked PPL
0.1	0.000	10.76
0.3	0.000	10.61
0.5	0.200	11.17
1.0	0.600	11.03
2.0	1.000	13.89

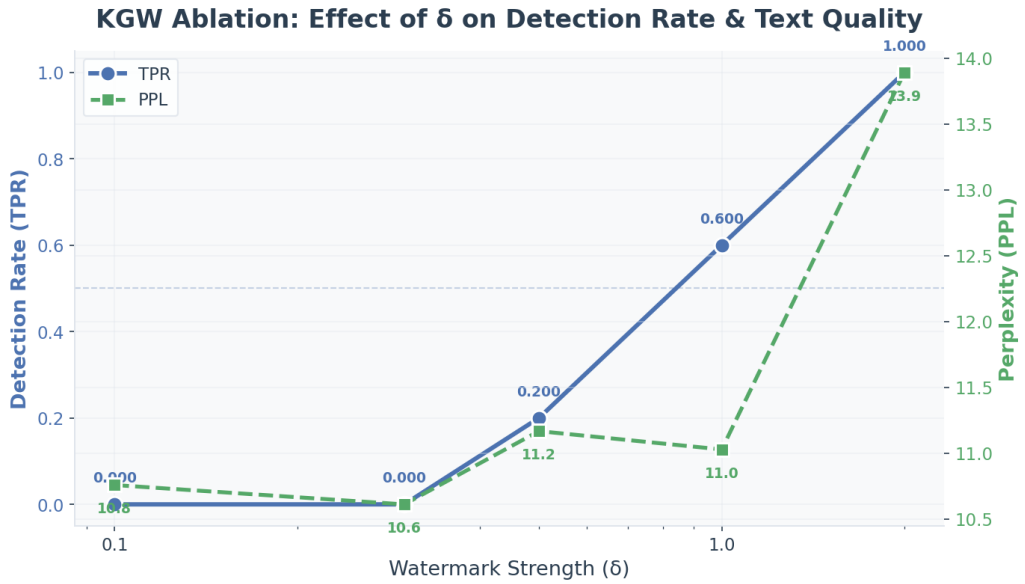


Figure 6: KGW ablation: effect of watermark strength δ on detection rate (TPR, left axis) and text perplexity (PPL, right axis). TPR increases monotonically while PPL rises sharply at $\delta = 2.0$.

0.650, and $\gamma = 0.75$ collapses to 0.000. The failure at $\gamma = 0.75$ stems from signal dilution: when nearly all vocabulary tokens are on the green list, the expected green-token proportion under the null hypothesis is close to 0.75 regardless of watermarking, so the excess z-score signal is negligible. PPL is stable across γ values (11.0–12.0), confirming that the green-list ratio is not the primary driver of quality degradation.

Table 6: KGW ablation on green-list ratio γ ($\delta = 1.0$ fixed, $n = 20$).

γ	TPR	Watermarked PPL
0.25	0.650	11.72
0.50	0.700	11.99
0.75	0.000	10.98

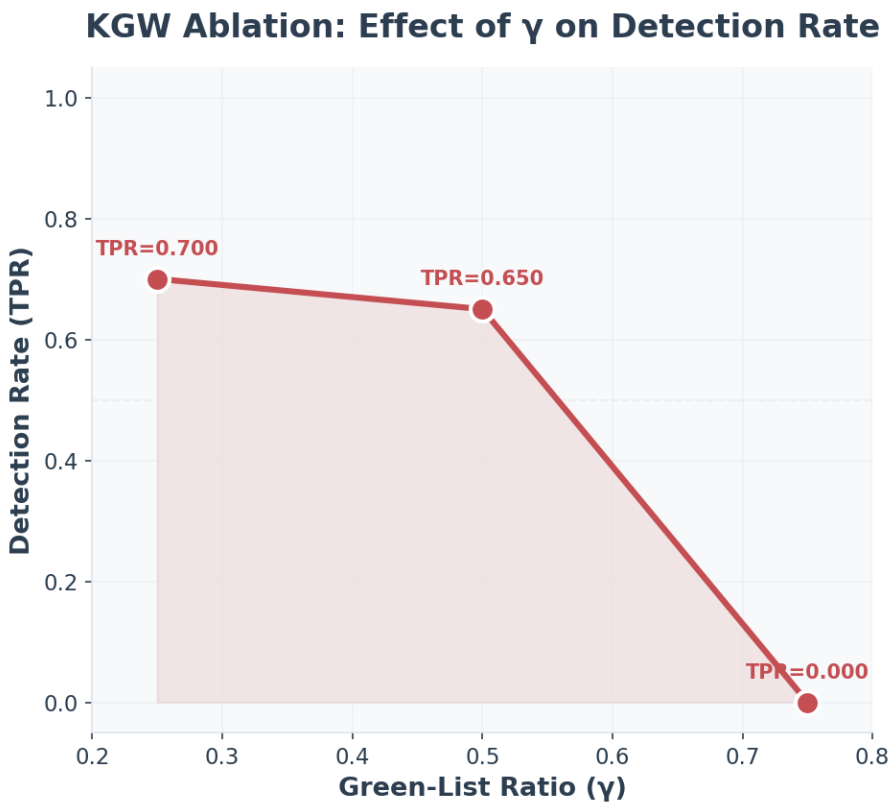


Figure 7: KGW ablation: effect of green-list ratio γ on detection rate (TPR). TPR collapses to zero at $\gamma = 0.75$ due to signal dilution.

4.4.3 EXP: Hyperparameter Sensitivity

We ablate two EXP hyperparameters independently. **Detection threshold** $\tau \in \{10^{-6}, 10^{-4}, 10^{-2}\}$ (prefix $k = 4$ fixed): TPR is 0.900 at $\tau = 10^{-6}$ and 0.950 at both $\tau = 10^{-4}$ and 10^{-2}, with FPR $= 0.000$ throughout (Figure 8). **Prefix length** $k \in \{1, 4, 8\}$ ($\tau = 10^{-6}$ fixed): TPR is 1.000 at $k = 1$ and $k = 8$, with a minor dip to 0.950 at $k = 4$, and FPR $= 0.000$ throughout (Figure 9). In both cases, variation across settings is negligible (≤ 0.1 TPR). This insensitivity stems from the extremely strong watermark signal embedded by EXP: watermarked p-values typically lie in the range 10^{-20}–10^{-103}, far below any reasonable threshold, making the exact parameter values inconsequential.

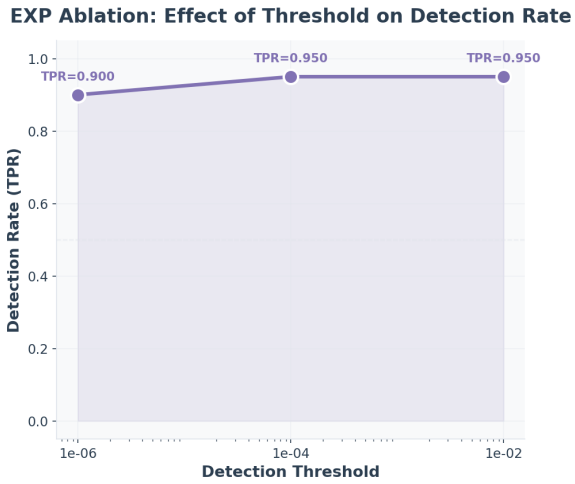


Figure 8: EXP ablation: effect of detection threshold τ on TPR. Performance is stable across three orders of magnitude.

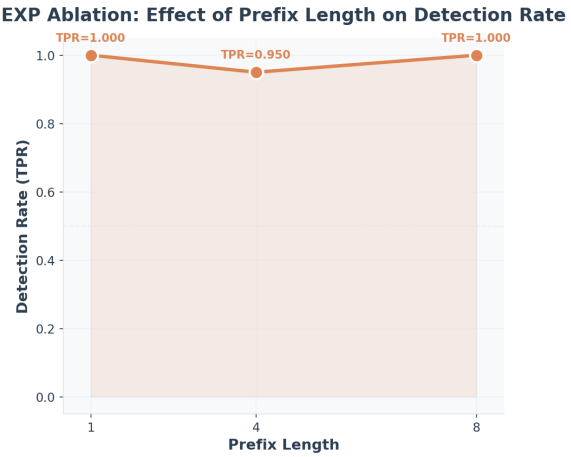


Figure 9: EXP ablation: effect of prefix length k on TPR. Minor variation (≤ 0.05) across all tested values.

4.5 Robustness Against Paraphrasing Attacks (KGW and EXP)

Table 7 and Figures 10–11 report TPR of KGW and EXP under two word-level attacks ($n = 20$, attack rates matched to the baseline experiment).

Table 7: TPR of EXP and KGW under word-level paraphrasing attacks ($n = 20$). FPR $= 0.000$ in all conditions.

Method	No Attack	Word-S	Word-D
EXP	1.000	0.050	0.000
KGW	0.800	0.050	0.100

Both methods suffer dramatic degradation under attack. Word-S reduces the TPR of both methods to 0.050, effectively destroying watermark detectability in 19 out of 20 cases. Under Word-D, EXP falls to 0.000 while KGW retains a marginal TPR of 0.100.

Both collapses share the same root cause: both watermarks are fundamentally token-level mechanisms whose signals depend on the exact identity of generated tokens. Word-S invalidates this by changing token identities — for KGW, synonym substitution alters both the substituted token's green/red identity and the context hash for subsequent tokens, breaking the statistical signal throughout; for EXP, it disrupts the accumulated score at each modified position. The marginal advantage of KGW under Word-D (0.100 vs. 0.000) arises because deletion preserves the green/red identity of surviving tokens, allowing a small number of high-confidence samples to still exceed the z-score threshold, whereas EXP's joint p-value across the full sequence is more sensitive to the reduction in sequence length that deletion causes.

4.6 Robustness of the Combined Method

To evaluate whether the hybrid KGW+EXP design improves robustness, we conduct a larger-scale robustness experiment ($n = 50$ prompts from C4, generation length 100 tokens) comparing all three methods under Word-S (20% substitution rate) and Word-D (30% deletion rate). Results are shown in Table 8 and Figure 12.

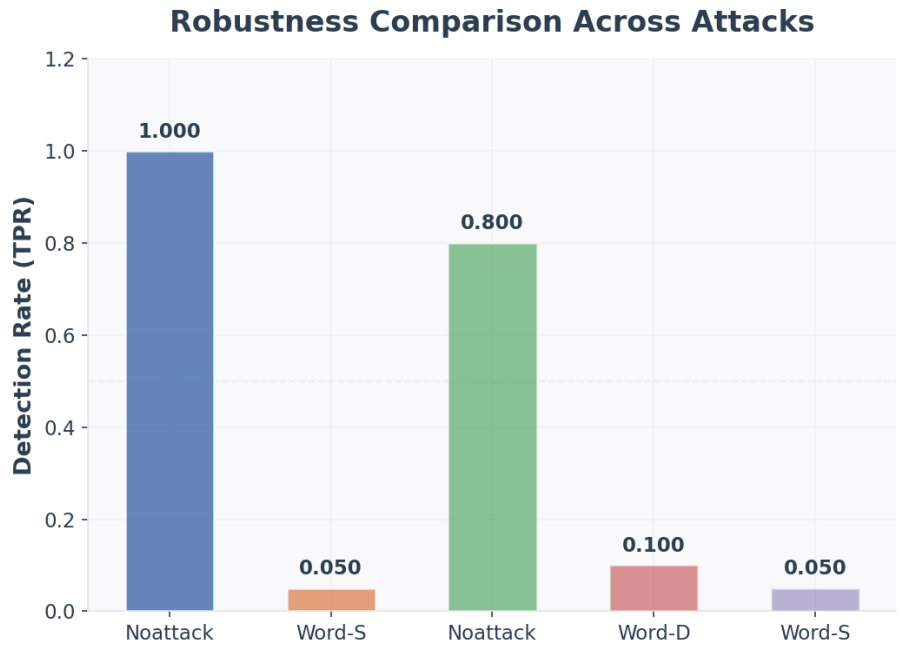


Figure 10: Robustness of EXP and KGW across attack conditions ($n = 20$). Both methods collapse under Word-S and Word-D attacks.

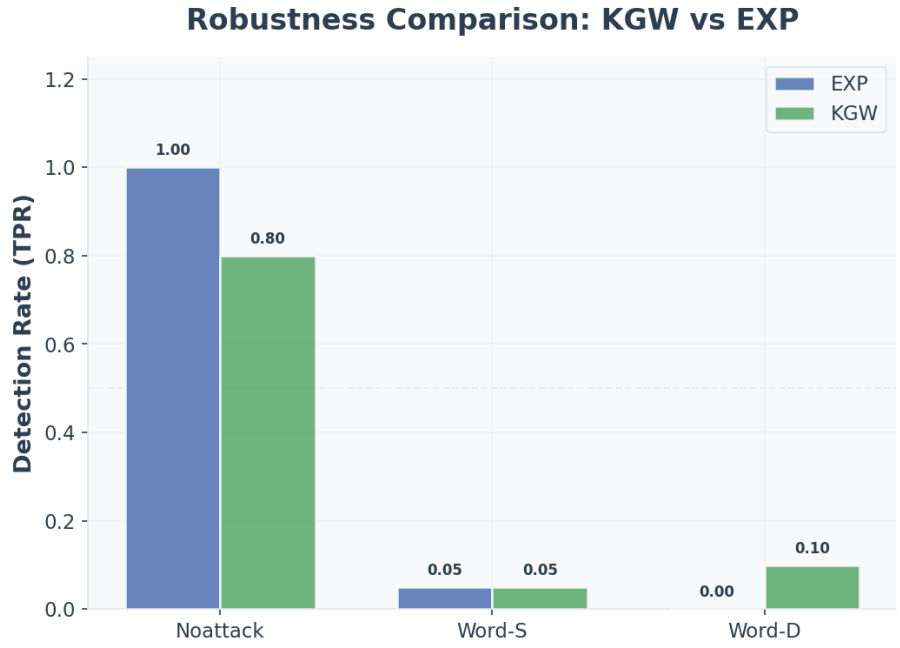


Figure 11: Side-by-side robustness comparison of EXP vs. KGW under no attack, Word-S, and Word-D conditions.

Table 8: Robustness comparison of KGW, EXP, and Combined ($n = 50$, C4 dataset). FPR $= 0.000$ in all conditions.

Method	**No Attack**	**Word-S (20%)**	**Word-D (30%)**
KGW	0.94	0.20	0.12
EXP	0.98	0.86	0.36
Combined	1.00	0.90	0.68

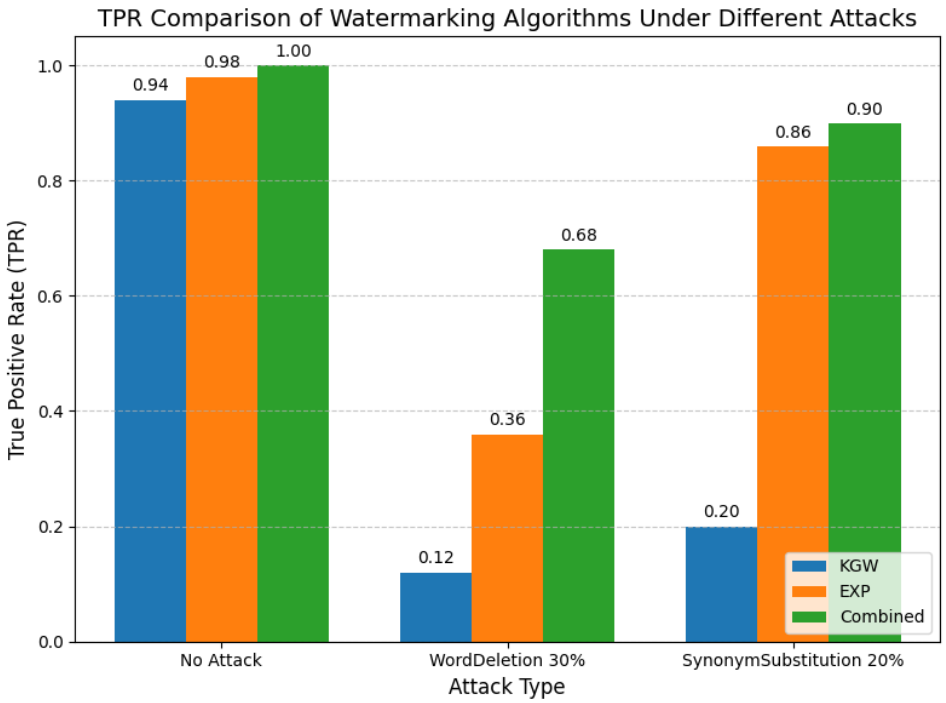


Figure 12: Robustness comparison of KGW, EXP, and Combined under no attack, Word-S (20%), and Word-D (30%) conditions ($n = 50$, C4 dataset).

The Combined method achieves TPR $= 1.000$ with no attack, and retains TPR $= 0.900$ under Word-S and TPR $= 0.680$ under Word-D — substantially outperforming both individual methods. Under Word-S, EXP alone achieves TPR $= 0.860$, which is already far superior to KGW (0.200); the Combined method gains a further 4 percentage points, suggesting that the Fisher fusion amplifies the residual EXP signal that survives synonym substitution. The improvement is most dramatic under Word-D: Combined (0.680) outperforms EXP (0.360) by 32 percentage points and KGW (0.120) by 56 percentage points. This is explained by the complementary nature of the two sub-detectors: the KGW z-score, computed over surviving tokens, contributes positive signal even after deletion, and the Fisher combination allows this residual signal to boost the combined statistic above the detection threshold in cases where neither detector alone would succeed.

These results demonstrate that the KGW+EXP hybrid design substantially addresses the robustness limitation identified in Section 4.5. While neither individual method provides reliable protection under word-level attacks, their statistical fusion via Fisher's method produces a watermark that is meaningfully harder to remove.

5 Evaluation

5.1 Overall Comparison

Table 9 consolidates the multi-dimensional comparison among all three methods. EXP dominates KGW on detection and text quality under benign conditions, while the Combined method achieves the best overall profile: perfect no-attack detection, near-baseline text quality, and substantially improved robustness over both individual methods.

Table 9: Multi-dimensional comparison of EXP, KGW, and Combined.

Dimension	EXP	KGW	Combined	Verdict
Detection (no attack)	TPR=1.00, FPR=0.00	TPR=0.80, FPR=0.00	TPR=1.00, FPR=0.00	EXP / Combined
Text quality (PPL)	11.17 (lower than baseline)	25.05 (higher than baseline)	17.26 ($\approx$ baseline)	Combined
Robustness: Word-S	TPR=0.86	TPR=0.20	TPR=0.90	Combined
Robustness: Word-D	TPR=0.36	TPR=0.12	TPR=0.68	Combined
Score separability	$p \approx 10^{-30}$ (wtm.)	z-score overlap	Fisher score $> \theta$	EXP / Combined
Hyperparam. stability	Low sensitivity	Sensitive to δ, γ	Moderate (two configs)	EXP
Interpretability	Lower	Higher	Moderate	KGW

5.2 Detection Accuracy

EXP and Combined both achieve perfect detection (TPR $= 1.000$, FPR $= 0.000$) under default no-attack conditions, while KGW reaches only TPR $= 0.800$. The p-value separation shown in Figure 4 is essentially perfect for EXP: all 20 watermarked sequences receive p-values below 10^{-5}, while no natural sequence falls below 10^{-2}. The exponential resampling mechanism ensures that each generated token contributes a statistically independent signal that accumulates multiplicatively, producing an aggregate p-value that grows exponentially with sequence length.

KGW's z-test-based detection is inherently noisier. The z-score is a linear statistic that averages the green-token excess over the sequence; at 200 tokens with the default threshold of 4.0, four samples fail to exceed the detection boundary (TPR $= 0.800$). The score distribution in Figure 3 shows that watermarked and natural z-scores partially overlap, a structural limitation of the additive mechanism at moderate δ. Raising δ to 2.0 achieves TPR $= 1.000$ but at a significant quality cost (PPL $= 13.89$), illustrating that KGW cannot simultaneously optimize both dimensions.

The Combined method resolves this tension: by fusing the two detection signals via Fisher's method, it achieves TPR $= 1.000$ without requiring an aggressive δ. All three methods achieve FPR $= 0.000$, confirming that false positive rates are well controlled across all configurations.

5.3 Text Quality (Stealth Analysis)

Table 4 and Figure 13 reveal starkly different quality profiles across the three methods. KGW (PPL $= 25.05$) is the only method that raises perplexity above the natural baseline (19.63), a direct consequence of the additive logit bias forcing generation toward the green-list subset and away from the model's preferred token distribution. EXP (PPL $= 11.17$) produces anomalously low perplexity, reflecting its Gumbel resampling mechanism which systematically selects high-probability tokens and results in more conservative, less diverse output — lower PPL here is not a quality improvement but a sign of reduced lexical richness.

The Combined method achieves PPL $= 17.26$, almost exactly matching the natural baseline (ΔPPL $= -2.37$). This near-neutral quality impact arises because the KGW and EXP biases partially cancel: KGW's green-list pressure pushes logits upward for a fixed token subset, while

EXP's resampling redistributes probability mass more broadly. The net effect is a generation distribution closer to the unperturbed model output. From a stealth perspective, Combined is the strongest of the three methods: its watermarked text is statistically indistinguishable from natural output by a perplexity-based detector, unlike KGW (significantly higher PPL) or EXP (suspiciously lower PPL).

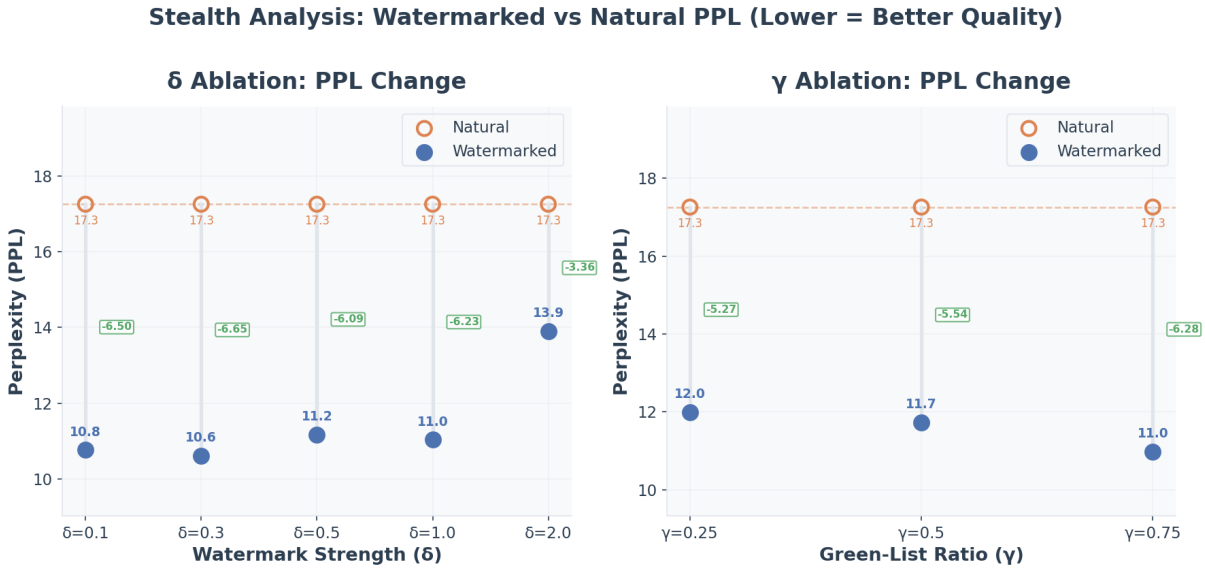


Figure 13: Stealth analysis: watermarked vs. natural PPL across KGW parameter settings. Left: δ ablation; right: γ ablation. The orange dashed line marks the natural text baseline (PPL $=$ 17.26 for the KGW/EXP baseline experiment; 19.63 for the Combined experiment using GPT-2).

5.4 Robustness

The robustness results reveal a clear performance hierarchy: Combined $>$ EXP $>$ KGW across both attack types, as shown in Table 8 and Figure 12.

KGW collapses most severely under both attacks (Word-S: 0.200, Word-D: 0.120), confirming that the green-list signal — which depends on exact token identities and context hashes — is highly fragile to any token-level perturbation. EXP performs substantially better under Word-S (TPR $=$ 0.860), because its resampling score is computed independently at each position and the surviving unmodified tokens still contribute strong signal to the aggregate p-value. However, EXP degrades more under Word-D (TPR $=$ 0.360) because its joint p-value is sensitive to sequence length reduction.

The Combined method outperforms both across the board. Under Word-S, it achieves TPR $=$ 0.900, a 4-point gain over EXP alone, suggesting that the residual KGW signal for unsubstituted tokens provides a modest additional boost via the Fisher combination. The largest gain is under Word-D: Combined achieves TPR $=$ 0.680 versus EXP's 0.360 and KGW's 0.120. This 32-point improvement over EXP arises from the complementary degradation profiles of the two sub-detectors: when deletion shortens the sequence and weakens EXP's joint p-value, the KGW z-score — computed only over surviving tokens — can still contribute meaningful signal, and the Fisher fusion lifts the combined statistic above the detection threshold in marginal cases where neither method alone would succeed.

These results demonstrate that the hybrid design directly addresses the robustness limitation of token-level watermarks. Although neither KGW nor EXP alone provides reliable protection under realistic word-level attacks, their statistical fusion substantially raises the bar for an adversary attempting removal.

5.5 Pros, Cons, and Deployment Recommendations

Based on our experiments, we summarize the practical deployment profile of each method.

EXP is the preferred single-method choice when detection reliability and text quality under benign conditions are the primary concerns. Its key advantages are: (1) perfect TPR and FPR $= 0$ at default settings; (2) low PPL (11.17), though this reflects reduced diversity rather than genuine quality improvement; and (3) near-zero sensitivity to hyperparameters. Its main limitation is vulnerability to word-level attacks, particularly Word-D (TPR $= 0.360$).

KGW may be preferred where interpretability and explicit parameter control are important. Its δ–γ parameter space provides a transparent detectability–quality trade-off. However, it achieves the weakest robustness of the three methods, and its PPL elevation (25.05) makes watermarked text noticeably less fluent. The catastrophic failure at $\gamma = 0.75$ (TPR $= 0.000$) is an important practical warning about misconfiguration risk.

Combined is the recommended method for adversarial deployment scenarios. It achieves perfect no-attack detection, near-baseline text quality (ΔPPL $= -2.37$), and the strongest robustness under both Word-S (0.900) and Word-D (0.680). The main trade-off is increased configuration complexity: two separate watermark configs plus Fisher fusion parameters must be maintained. For IP protection use cases where paraphrasing attacks are a realistic threat, Combined's robustness advantage decisively outweighs this overhead.

6 Conclusion

In this paper, we reproduced and systematically evaluated two representative LLM watermarking methods — KGW and EXP — and proposed a Combined hybrid method that fuses both via Fisher's statistical combination. Experiments were conducted on text generated by Qwen2.5-1.5B-Instruct using the MarkLLM toolkit, covering three core desiderata: detection accuracy (TPR and FPR), text quality (PPL evaluated by GPT-2), and robustness against word-level paraphrasing attacks.

Our experiments yield four principal findings. First, under benign conditions EXP is strictly superior to KGW as a standalone method: EXP achieves perfect detection (TPR $= 1.000$, FPR $= 0.000$) with a watermarked PPL of 11.17, versus KGW's TPR $= 0.800$ and PPL $= 25.05$. KGW's interpretable green-list mechanism comes at a real cost — its detection reliability depends heavily on the choice of δ and γ, and misconfiguration (e.g., $\gamma = 0.75$) causes complete detection failure.

Second, both KGW and EXP are highly vulnerable to word-level attacks when used alone. Under Word-S (20% substitution), KGW drops to TPR $= 0.200$ and EXP to 0.860; under Word-D (30% deletion), KGW drops to 0.120 and EXP to 0.360. This exposes a fundamental limitation of generation-time token-level watermarks: the embedded signal depends on the exact identity of generated tokens, and any operation that alters token identities — even while preserving meaning — degrades detectability.

Third, the proposed Combined method substantially addresses this robustness limitation. By jointly embedding KGW and EXP signals during generation and fusing their detection statistics via Fisher's method, Combined achieves TPR $= 1.000$ with no attack, TPR $= 0.900$ under

Word-S, and TPR $= 0.680$ under Word-D — outperforming EXP by 4 and 32 percentage points respectively under the two attacks. The Fisher fusion exploits the complementary degradation profiles of the two sub-detectors: when synonym substitution weakens KGW's context-hash-based signal, EXP's position-independent resampling score compensates; when deletion shortens the sequence and weakens EXP's joint p-value, the KGW z-score over surviving tokens provides residual uplift.

Fourth, Combined achieves near-baseline text quality (PPL $= 17.26$, ΔPPL $= -2.37$ relative to the natural baseline of 19.63), making it the most stealthy of the three methods. KGW's additive logit bias raises PPL noticeably above baseline, while EXP's resampling anomalously suppresses PPL below it — both deviations are potentially detectable by a perplexity-based adversary. Combined's near-neutral quality impact makes its watermarked text statistically closest to natural LLM output.

From this project, we learned that the central challenge in LLM watermarking is not *detection* under ideal conditions — all three methods achieve zero false positives, and both EXP and Combined achieve perfect TPR without attack — but *robustness* under adversarial perturbation. A watermark that is easily removed by a basic synonym substitution tool offers limited practical value for IP protection. The Combined method demonstrates that statistical fusion of complementary token-level signals can meaningfully raise the robustness bar without sacrificing detection accuracy or text quality. Nevertheless, even Combined's TPR of 0.680 under Word-D leaves room for improvement, and future work should explore semantic-level watermarking approaches that encode signals in higher-level properties of text inherently preserved under meaning-preserving transformations.

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